



SentinelAI

*AI-Powered Multi-Modal Digital Threat Detection & Digital Authenticity
Analysis Platform*

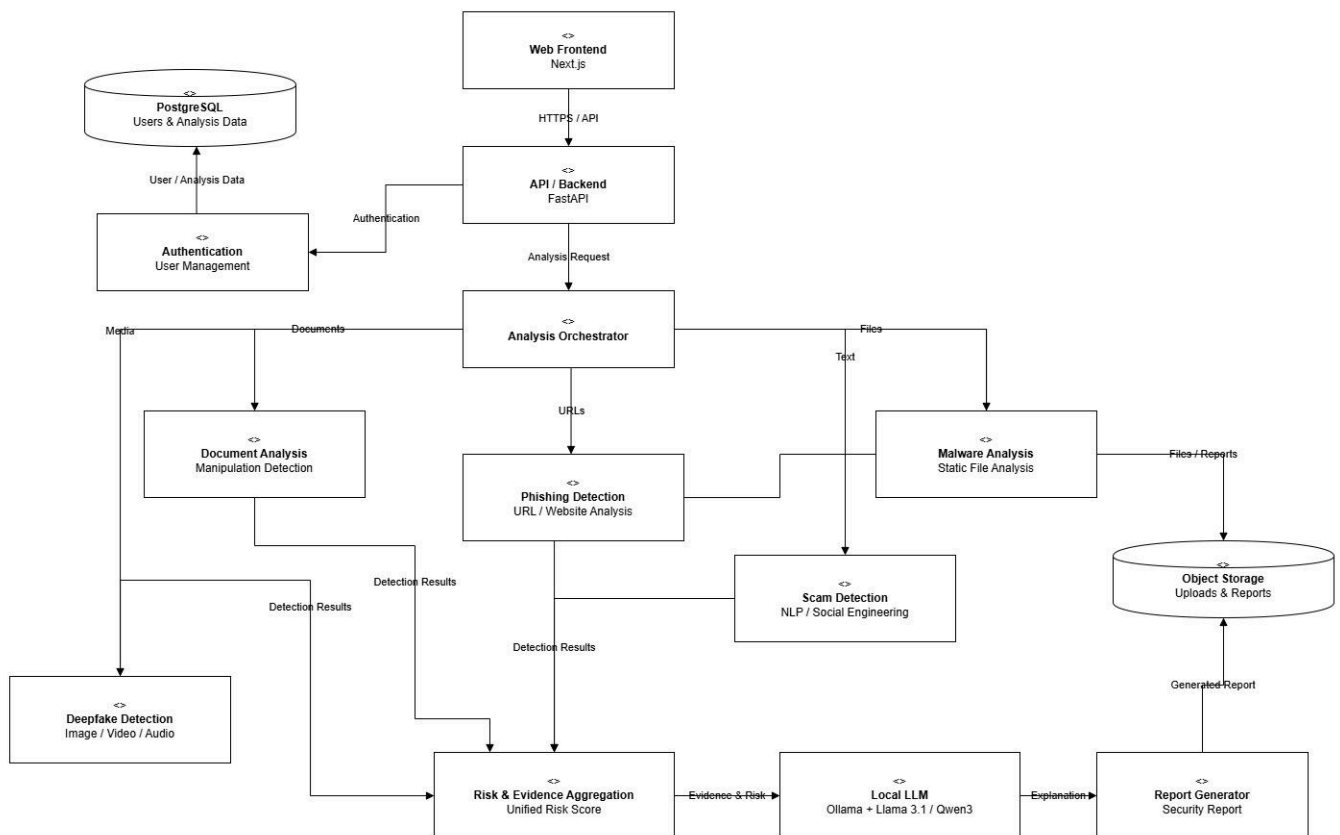
Software Project Design and Development CSE 416

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Component Diagram

The Component diagram maps out the structural architecture of the system, showing how different modules connect and communicate.

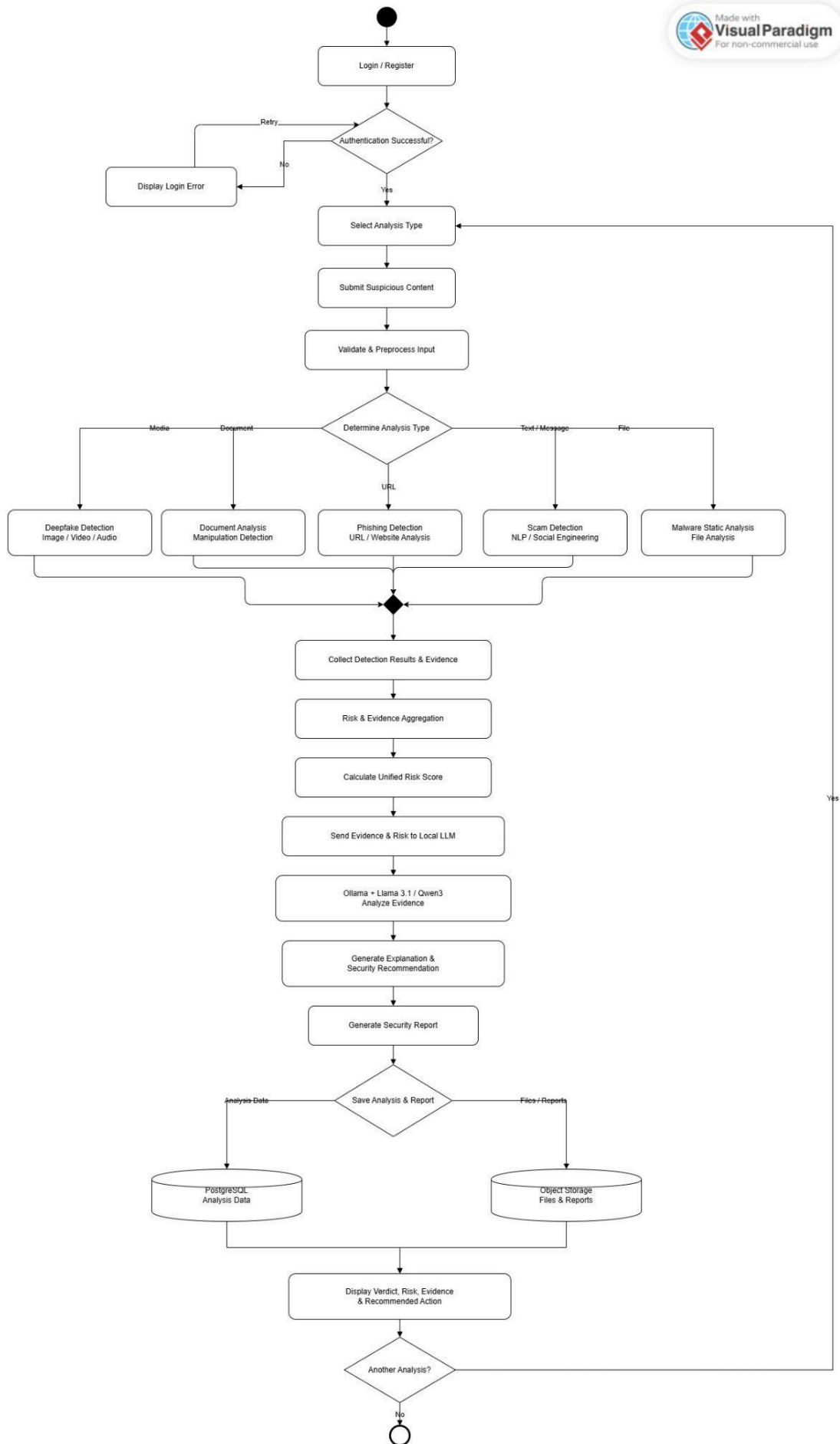
- Frontend and Backend: A "Web Frontend" built with Next.js communicates via HTTPS/API with an "API / Backend" built with FastAPI. The backend interacts with an Authentication module that manages users in a PostgreSQL database.
- Orchestration and Detection Engines: The API sends analysis requests to the "Analysis Orchestrator," which routes the data to the appropriate detection module: Document Analysis, Deepfake Detection, Phishing Detection, Scam Detection, or Malware Analysis.
- Aggregation and AI: The detection modules send their results to a "Risk & Evidence Aggregation" component to formulate a Unified Risk Score. This aggregated data is passed to a "Local LLM" (Ollama + Llama 3.1 / Qwen3) to generate explanations.
- Output: The LLM's output goes to a "Report Generator" that creates a security report and saves it to "Object Storage".



Activity Diagram

The Activity Diagram illustrates the step-by-step workflow a user takes when interacting with the system, as well as the internal processing steps.

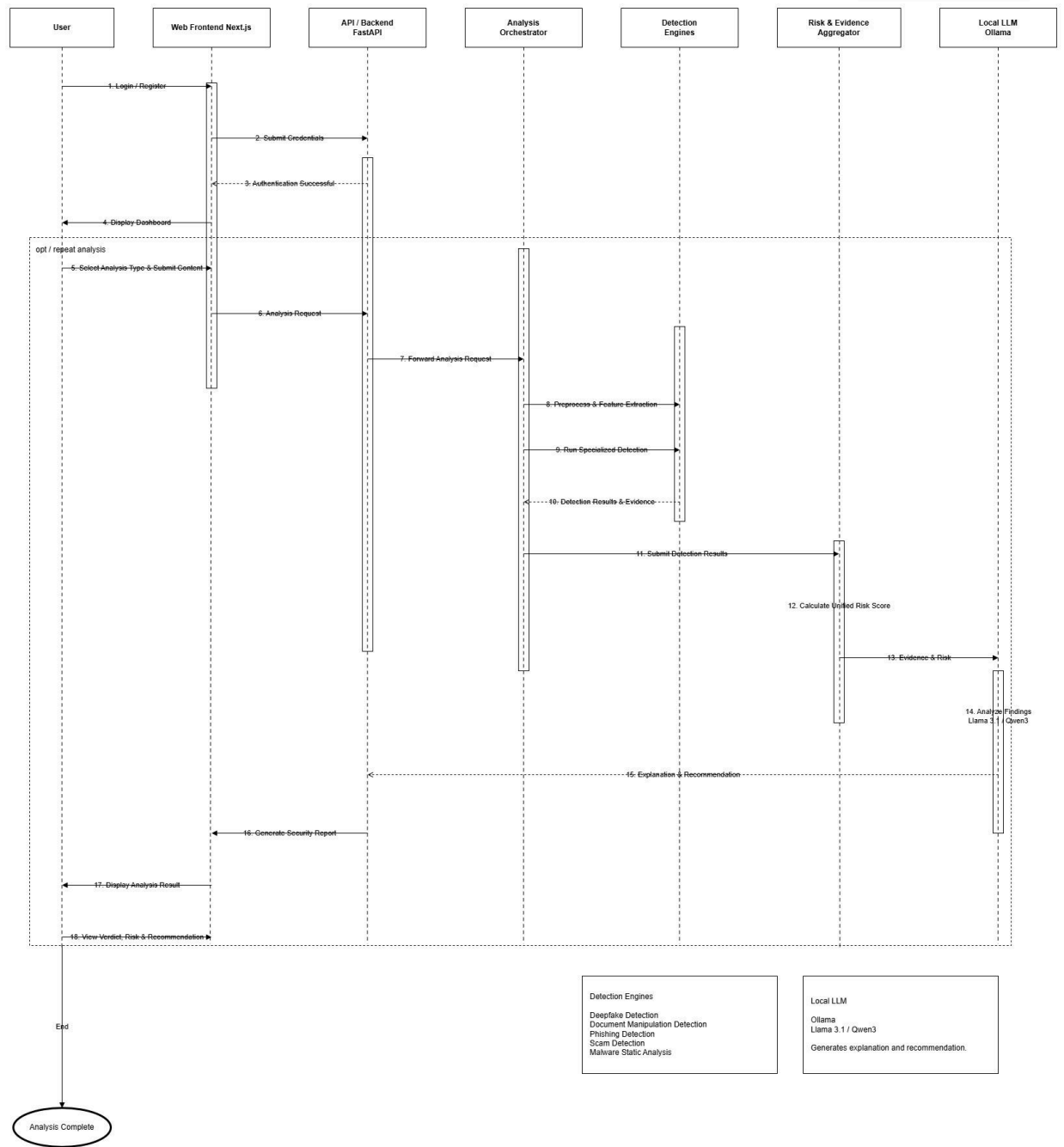
- **Authentication:** The process begins with the user attempting to log in or register, looping back to display an error if authentication fails.
- **Submission and Routing:** Once authenticated, the user selects an analysis type and submits suspicious content, which the system then validates and preprocesses. The workflow branches based on the input type into five specific engines: Deepfake Detection (Media), Document Analysis (Document), Phishing Detection (URL), Scam Detection (Text/Message), or Malware Static Analysis (File).
- **Aggregation and AI Analysis:** After the specialized engine runs, the system collects the results, aggregates the risk and evidence, and calculates a Unified Risk Score. This evidence and risk data is then sent to a Local LLM (using Ollama + Llama 3.1 / Qwen3) to generate an explanation and a security recommendation.
- **Reporting and Storage:** Finally, a security report is generated, saved to PostgreSQL (for analysis data) and Object Storage (for files/reports), and the final verdict is displayed to the user, who can then choose to start another analysis.



Sequence Diagram

The Sequence diagram visualizes the chronological flow of messages and interactions between the system's actors and components over time.

- **User Flow:** It tracks the user's journey from logging in through the Web Frontend (Next.js) to submitting credentials to the API (FastAPI), receiving successful authentication, and viewing the dashboard.
- **Analysis Execution:** When a user submits content, the Web Frontend sends an Analysis Request to the API, which forwards it to the Analysis Orchestrator. The Orchestrator handles preprocessing, feature extraction, and runs specialized detection via the Detection Engines.
- **Risk Scoring and LLM Integration:** Detection results are submitted to the Risk & Evidence Aggregator to calculate a Unified Risk Score. The evidence and risk are then forwarded to the Local LLM (Ollama) to analyze findings and return an explanation and recommendation.
- **Final Output:** The backend generates a security report and sends the final analysis result back to the frontend, which displays the verdict, risk, and recommendation to the user, concluding the process.



Fishbone Diagram

The **Fishbone.png** diagram (also known as an Ishikawa or cause-and-effect diagram) identifies the root causes contributing to a central problem: the "Difficulty in Detecting and Assessing Digital Threats". The causes are grouped into six main categories:

- **Technology:** Driven by sophisticated deepfakes and rapidly evolving Generative AI.
- **Data:** Caused by limited labeled datasets and dataset bias and imbalance.
- **Detection:** Complicated by multiple threat categories and issues with false positives / false negatives.
- **Security:** Vulnerabilities stemming from malicious file handling and untrusted user input.
- **Usability:** Hindered by fragmented security tools and complex technical results.
- **Infrastructure:** Challenged by high compute requirements and large media processing overhead.

